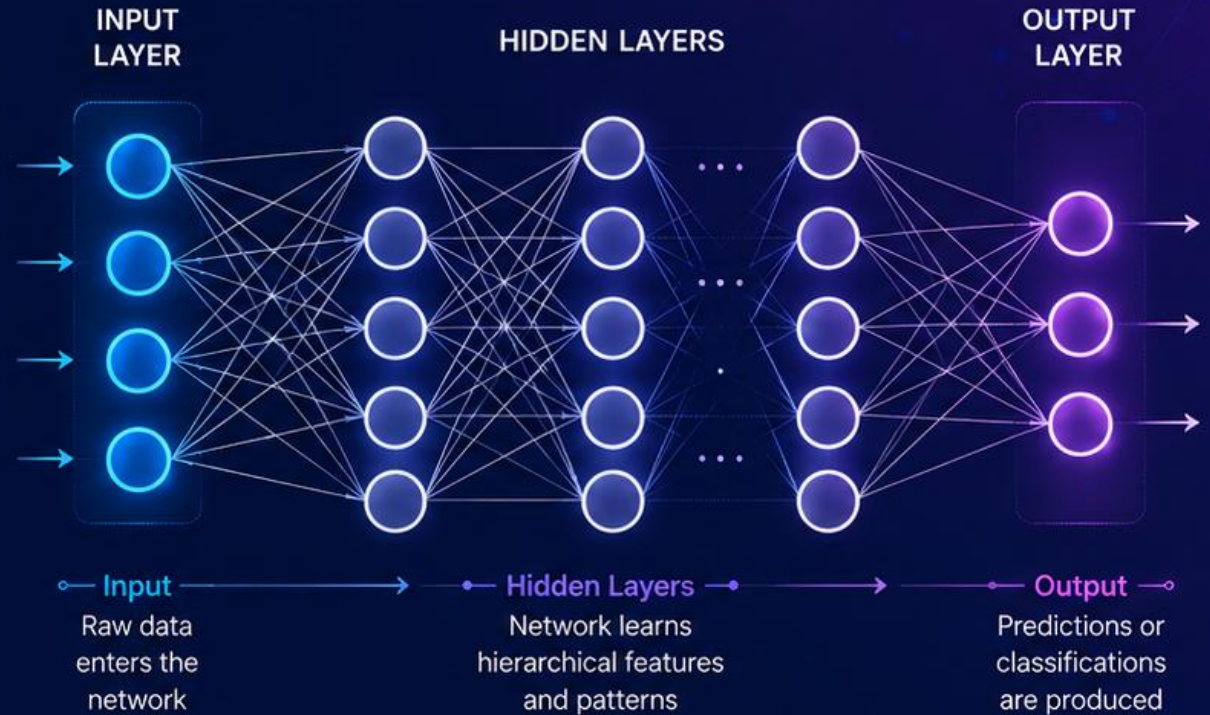




What is Deep Learning?

- ✓ Deep Learning is a branch of **Machine Learning**.
- ✓ It uses **multi-layered neural networks** to learn patterns from data.
- ✓ It automatically learns **features** from raw input.
- ✓ It is widely used in **images, speech, text,** and **medical diagnosis**.



WIDELY USED IN



Images



Speech



Text



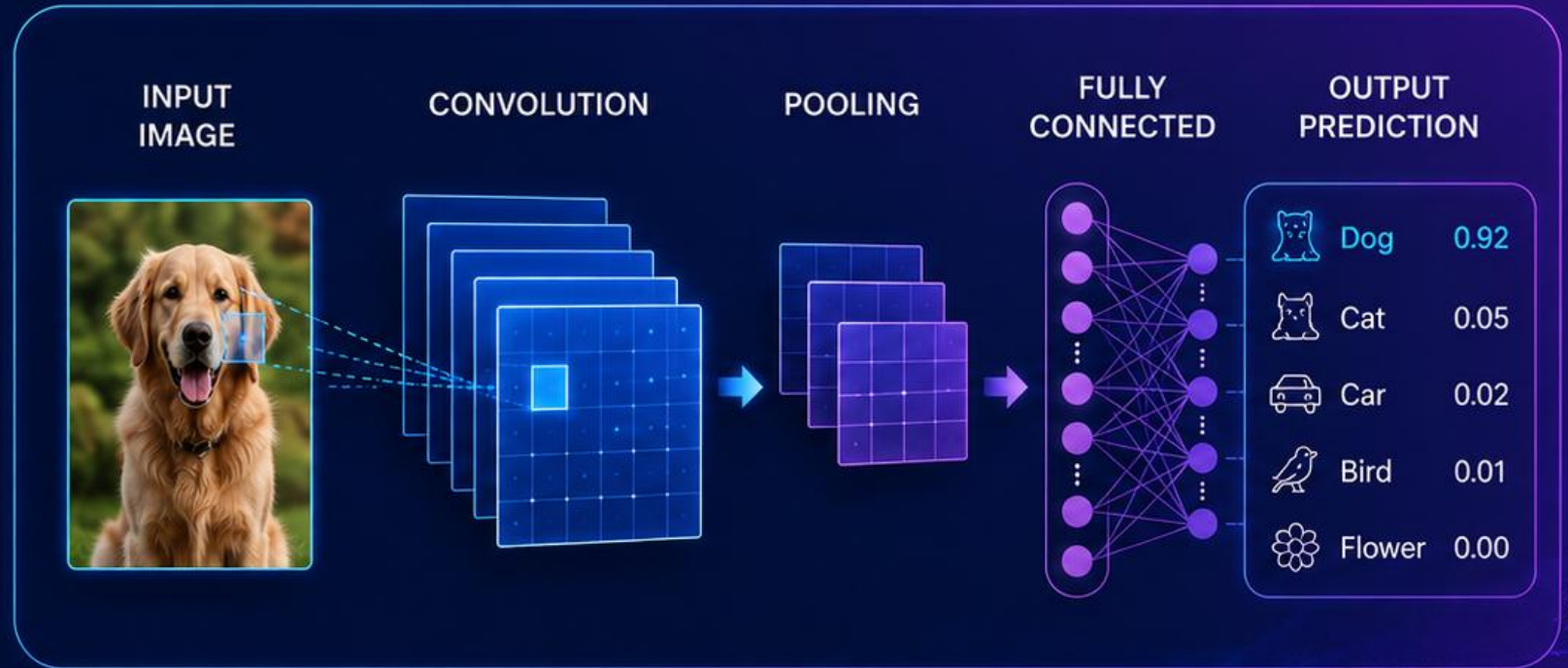
Healthcare





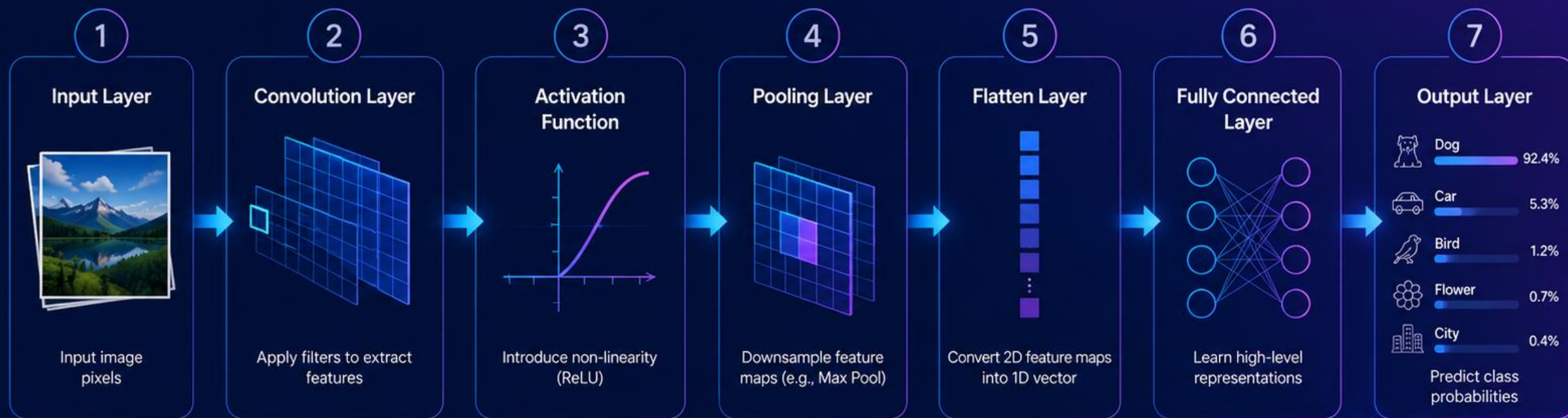
What is CNN?

- CNN stands for Convolutional Neural Network.
- It is a deep learning model mainly used for image-related tasks.
- CNN automatically extracts important visual features.
- It is highly effective for image classification.





CNN Architecture





Convolution Layer



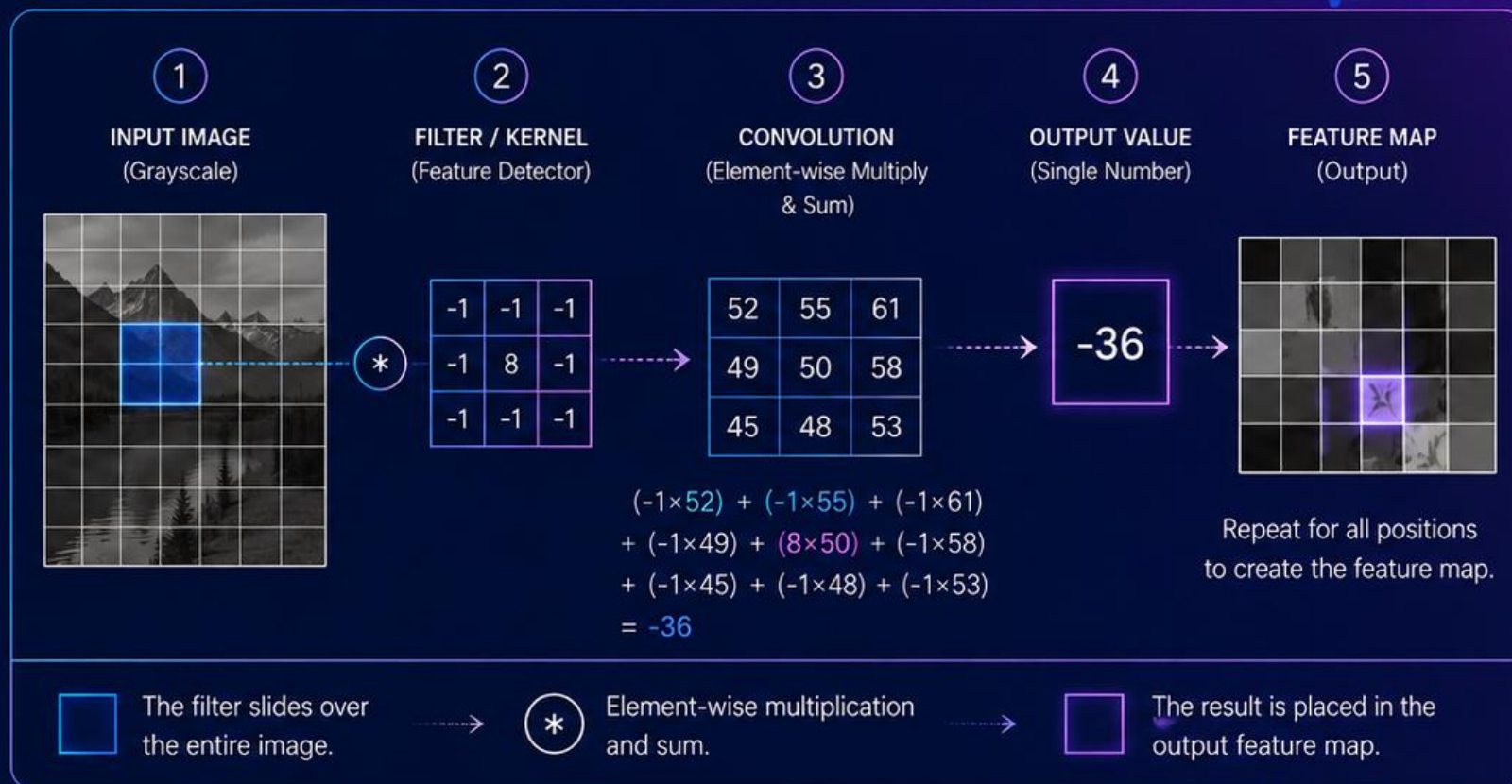
The convolution layer applies **filters or kernels** to the input image.



These filters detect features such as **edges, corners**, and **patterns**.



Different filters learn different **visual characteristics**.

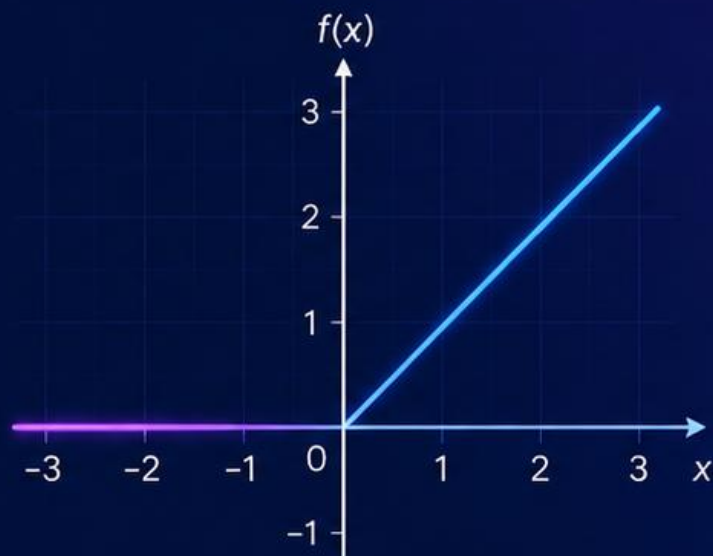




Activation Function

- ✓ The most common activation function in CNNs is ReLU.
- ✓ ReLU introduces non-linearity into the model.
- ✓ It helps the network learn complex patterns.
- ✓ It also improves training speed.

ReLU Function



$$f(x) = \max(0, x)$$

$$f(x) = 0, \text{ if } x \leq 0$$

$$f(x) = x, \text{ if } x > 0$$

Activation in Action

Feature Values (Before ReLU)		After ReLU (Activation)
-2.5	→	0.0
-1.0	→	0.0
0.0	→	0.0
1.2	→	1.2
3.4	→	3.4





Pooling Layer



Pooling reduces the spatial size of feature maps.



It keeps important information while reducing computation.



Common types are
Max Pooling and Average Pooling.

INPUT FEATURE MAP

1	3	2	4
5	6	7	8
2	1	0	3
4	2	1	0

MAX
POOLING
(2x2)

6	8
4	3

Takes the maximum value from each 2x2 region.

AVERAGE
POOLING
(2x2)

3.75	5.25
2.25	1.00

Takes the average value from each 2x2 region.



Pooling helps the model focus on the most important features by shrinking the feature maps while preserving key information.



Flatten and Fully Connected Layer



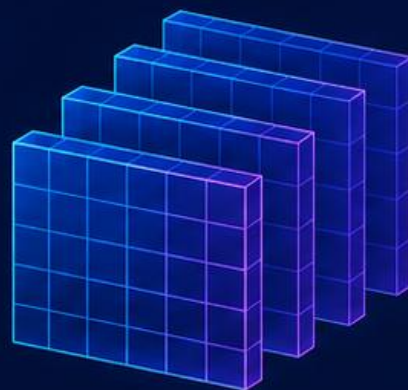
The flatten layer converts feature maps into a one-dimensional vector.



The fully connected layer uses these features for final decision making.



The output layer produces the final class prediction.



FEATURE MAPS
(From Convolutional
and Pooling Layers)



FLATTEN LAYER
Converts 3D feature maps
into a 1D vector



FULLY CONNECTED LAYER
Learns high-level patterns
for classification

OUTPUT LAYER Final Class Prediction		
	Dog	92.4%
	Car	5.3%
	Bird	1.2%
	Flower	0.7%
	City	0.4%



Dataset Preparation



Collect Images

Gather images from various sources.



Resize Images

Resize all images to a fixed dimension.



Normalize

Scale pixel values, usually between 0 and 1.



Split Data

Divide into training, validation, and testing sets.



Data Augmentation

Apply transformations such as rotation, flipping, zoom, etc.

RAW IMAGE COLLECTION



PREPROCESSING PIPELINE



Resize

Fixed dimension
(e.g., 224×224)



Normalize

Scale pixels to
0 – 1



Augment

Rotation, flip,
zoom, brightness, etc.



Shuffle

Randomize order
of samples

DATASET SPLIT



70%

TRAINING



15%

VALIDATION



15%

TESTING





Training Process



1

Input Images

Images are fed into the CNN model.



2

Forward Pass

The model extracts features and makes a prediction.



3

Compute Loss

Compare prediction with the actual label.



4

Backpropagation

Update model weights using gradients.



5

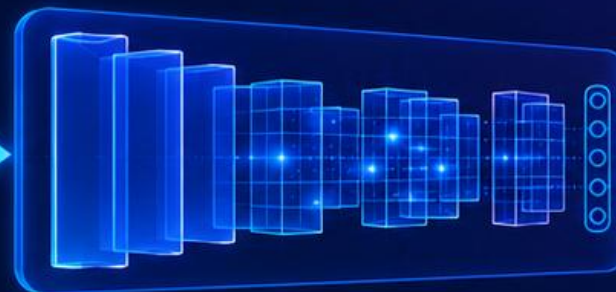
Repeat

Continue for many epochs until loss is minimized.

INPUT IMAGE



CNN MODEL



PREDICTED LABEL



ACTUAL LABEL



LOSS

Cross-Entropy Loss

$L = 0.15$

GRADIENTS (BACKPROPAGATION)





Evaluation Metrics



Accuracy

Percentage of correctly classified predictions.



Precision

Of predicted positives, how many are correct.



Recall

Of actual positives, how many are identified.



F1-score

Harmonic mean of Precision and Recall.



Confusion Matrix

Table showing TP, FP, TN, and FN.

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP True Positive Model predicted Positive and it was Positive.	FP False Positive Model predicted Positive but it was Negative.
	Negative	FN False Negative Model predicted Negative but it was Positive.	TN True Negative Model predicted Negative and it was Negative.



Accuracy



Precision



Recall



F1-score



Confusion Matrix





Applications



Medical Image Diagnosis

Detect diseases from X-rays, MRI, and CT scans.



Face Recognition

Identify and verify individuals.



Object Detection

Detect and classify objects in images.



Self-driving Cars

Recognize vehicles, pedestrians, and traffic signs.



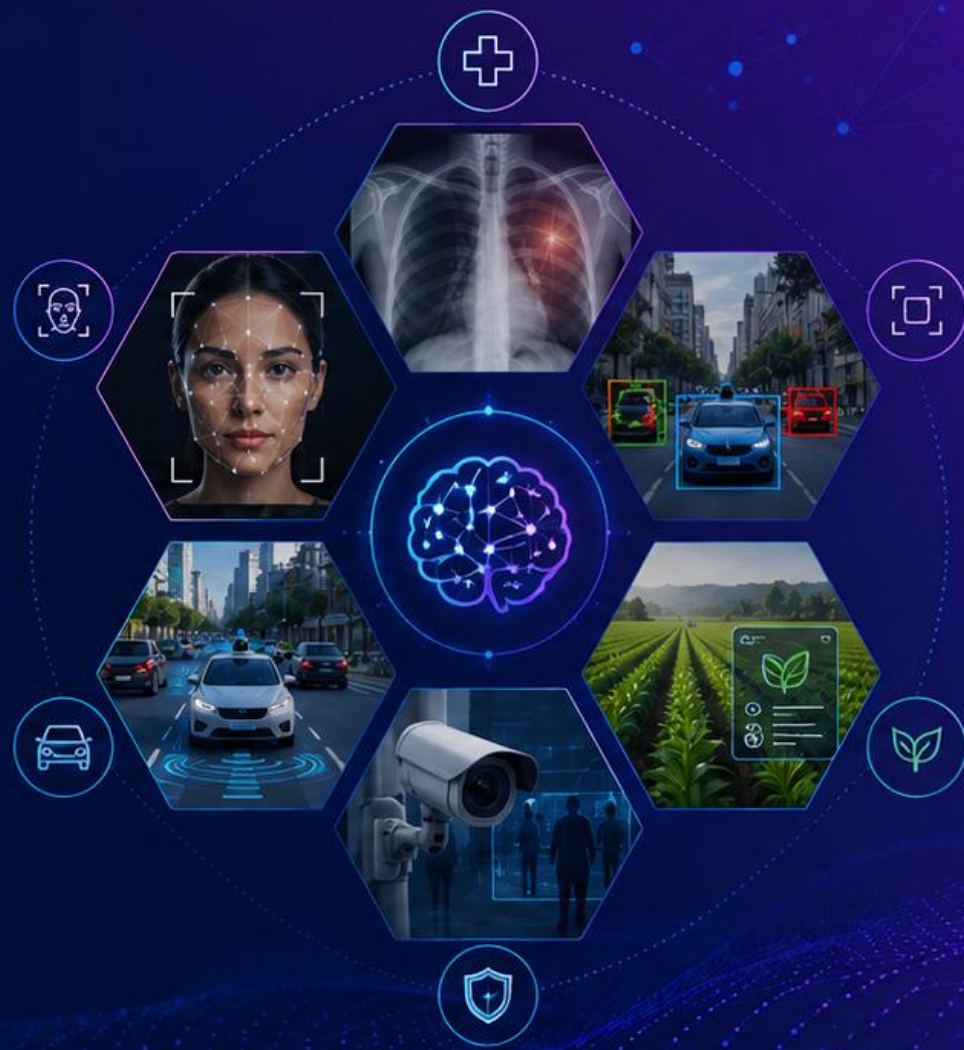
Agriculture

Detect crop diseases and classify plant types.



Security Systems

Surveillance, intrusion detection, and activity recognition.





Advantages



High Accuracy

Often achieves excellent performance.



Automatic Feature Extraction

Learns important features without manual engineering.



Works Well with Large Datasets

Improves as more data becomes available.



End-to-End Learning

Learns directly from raw images to final classification.



Useful in Real-world Problems

Applies to healthcare, automotive, agriculture, security, and more.





Challenges



Requires Large Datasets

Needs many labeled images for strong performance.



High Computational Power

Training requires GPUs and time.



Risk of Overfitting

Model may perform poorly on unseen data if not generalized well.



Training Takes Time

Deep models may need many epochs.



Difficult to Interpret

CNNs can behave like black boxes.

ACCURATE
& ROBUST
MODEL



Difficult to Interpret
Black box
predictions



Training Takes Time
Many epochs,
longer convergence



Risk of Overfitting
Poor generalization
if not careful



High Computational
Power
Expensive and
resource intensive



Large Datasets
Hard to collect
and label





Conclusion



Deep Learning with CNN has revolutionized image classification by automatically learning **powerful features** from images and delivering **high accuracy**.



It is a powerful tool with vast applications across multiple domains and will continue to **drive innovation** in the future.



HEALTHCARE



AUTOMOTIVE



AGRICULTURE



SECURITY



RETAIL



MANUFACTURING



EDUCATION





ADVANCING INTELLIGENCE.
SHAPING TOMORROW.

Thank You Questions?



Thank you for your attention.



DEEP LEARNING & COMPUTER VISION